



National University of Computer and Emerging Sciences



OS Lab Project Phase 03 Robo OS

Project Team

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1. Introduction of Robo OS

Robo OS: The Future of AI-Powered Computing

Welcome to the world of Robo OS, a revolutionary operating system built on the foundation of artificial intelligence. Robo OS is designed to push the boundaries of computing, offering an intuitive and intelligent experience for users of all levels.

Key Features:

- **AI-powered Interface:** Robo OS learns your preferences and adapts to your needs, making it easier than ever to navigate and complete tasks.
- **Personalized Experience:** Get recommendations, automate tasks, and enjoy an interface that evolves with you.
- **Enhanced Security:** AI-powered threat detection and prevention keep your data safe from cyberattacks.
- **Seamless Integration:** Works with your existing hardware and software for a smooth transition.
- **Constant Evolution:** Robo OS is always learning and improving, ensuring you have the latest features and functionalities.



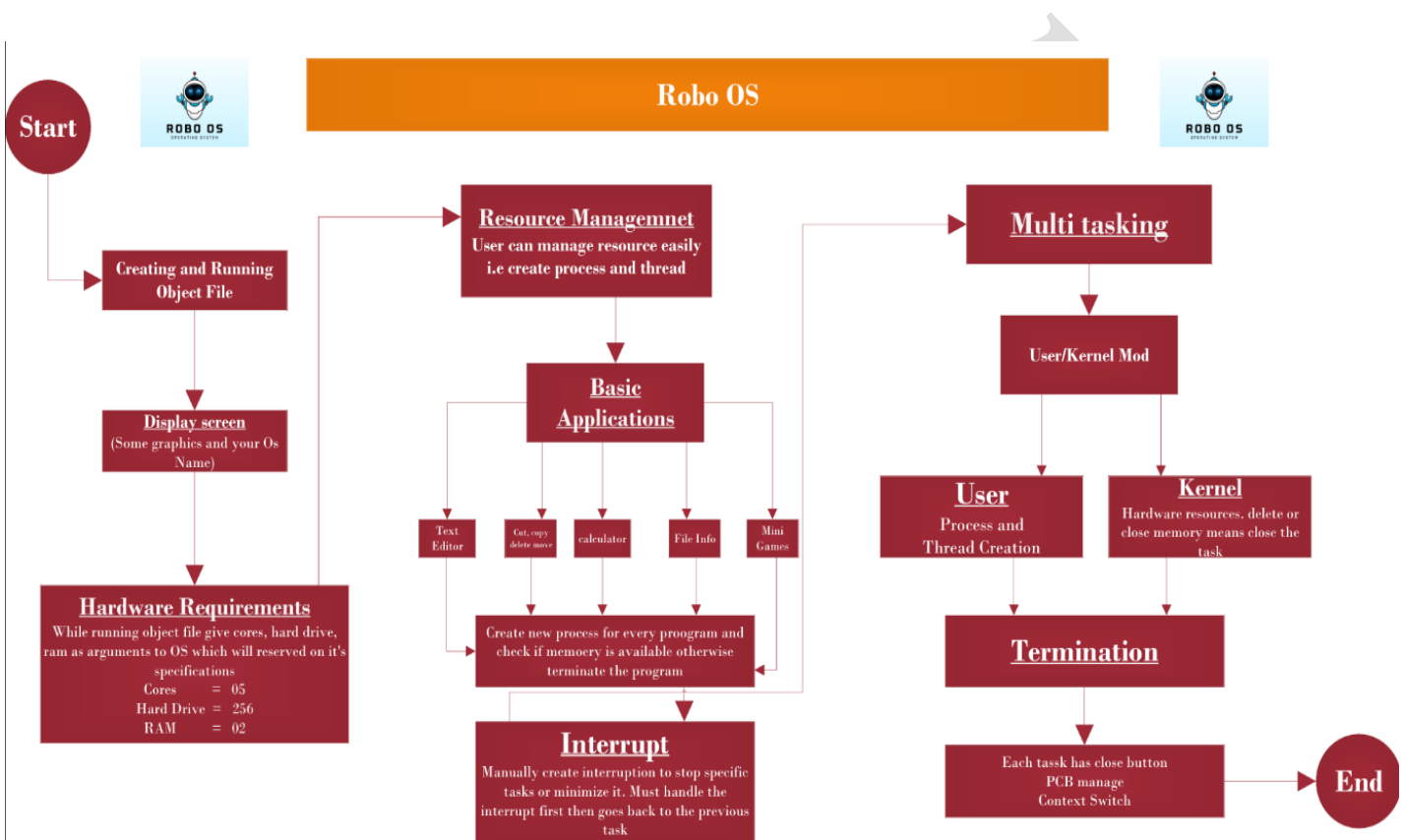


2. Objectives

- **Boost Productivity:** AI-powered automation and personalized recommendations for efficient workflows.
- **Personalization:** Adapts to individual needs and preferences for a truly personal experience.
- **Enhanced Security:** AI-powered threat detection and prevention for secure data protection.
- **Seamless Integration:** Works with existing hardware and software for a smooth transition.
- **Constant Evolution:** AI-driven continuous improvement and new features.
- **Democratize AI:** Accessible to all, empowering everyone with AI-powered computing.
- **Thriving Community:** Fosters collaboration and innovation among users and developers.
- **Shape the Future:** Pioneering AI in operating systems, setting new standards for computing.



3. Flowchart



4. Functionalities

- Process Management
- Memory Management
- File System Management
- Device Management
- User Interface



- Networking and Communication
- Security
- Error Handling and Fault Tolerance
- System Utilities
- Multitasking and Multithreading

5. Concept

- Message passing
- Process Creation
- Threading
- Multi-Tasking
- Multi-Threading
- Process control block (PCB)
- Memory Allocation/Deallocation
- Application/ Termination

6. Medium Blog

<https://medium.com/@metheBilalAshiq/robo-os-operating-system-project-5b8267c85731>



7. Final Output Screenshots

Starting

```
methebilalashiq@bilalashiq: ~/Robo OS
methebilalashiq@bilalashiq:~/Robo OS$ g++ -o final final.cpp -lsfml-graphics -lsfml-window -lsfml-system
methebilalashiq@bilalashiq:~/Robo OS$ ./final
```

```
methebilalashiq@bilalashiq: ~/Robo OS
Enter the amount of RAM (GB): 16
Enter the number of CPU cores: 8
```

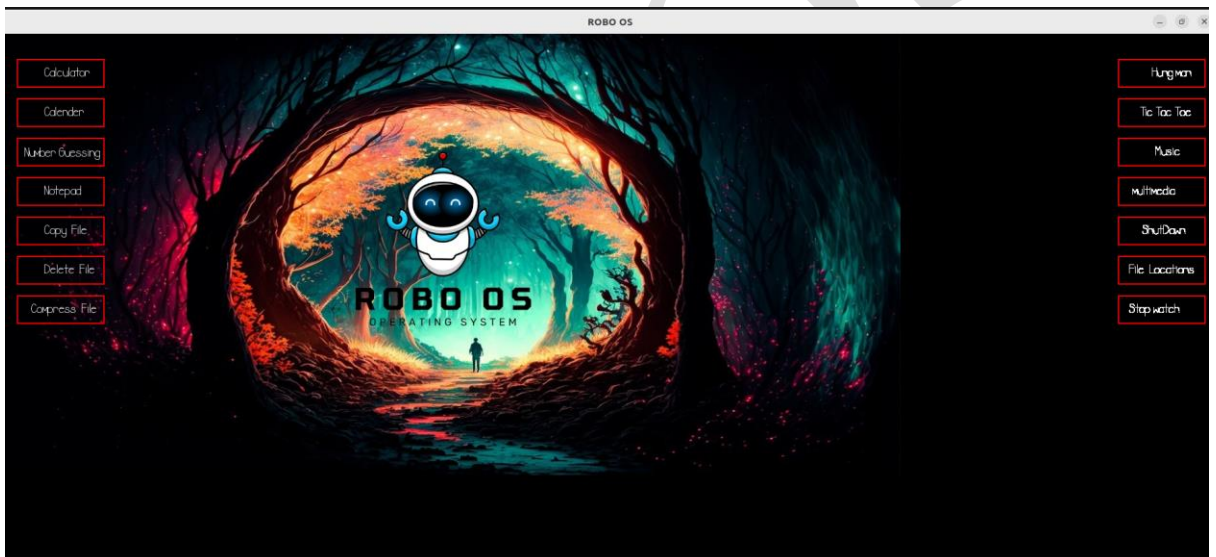
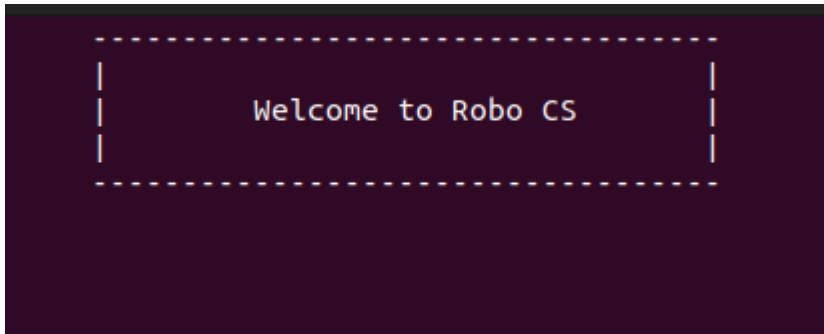
```
methebilalashiq@bilalashiq: ~/Robo OS

Please wait while window is loading

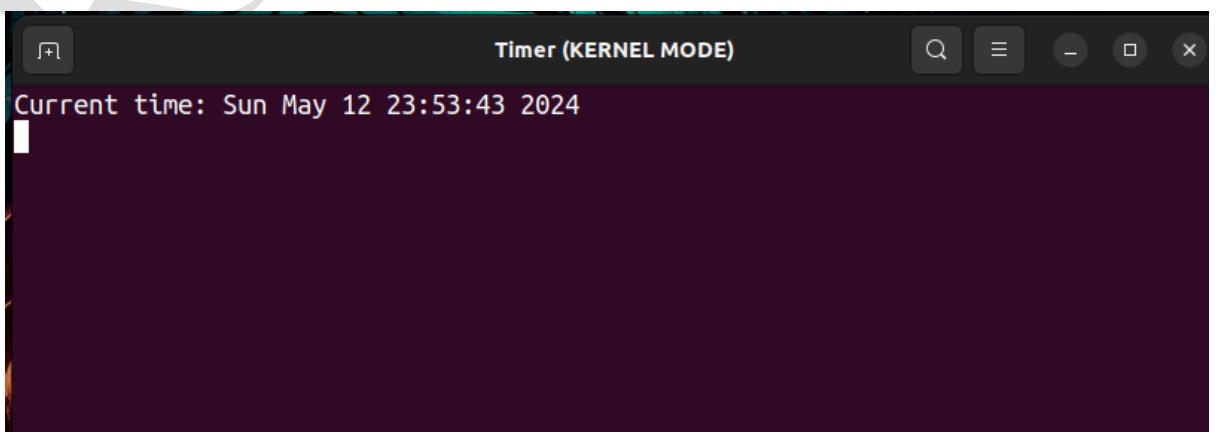
----- >>> ATTEMPTING TO LOGIN <<< -----
----- >>> GENERATING LOG FILES <<< -----
----- >>> ETING USER DATA <<< -----
----- >>> LOADING THE DIRECTORIES <<< -----
----- >>> CONNECTING TO THE MAIN OPERATING SYSTEM <<< -----
```



Display



Time





Process Runing

```
System Monitor (KERNEL MODE)

List of running processes:
systemd
kthreadd
rcu_gp
rcu_par_gp
slub_flushwq
netns
mm_percpu_wq
rcu_tasks_kthread
rcu_tasks_rude_kthread
rcu_tasks_trace_kthread
ksoftirqd/0
rcu_preempt
migration/0
idle_inject/0
cpuhp/0
cpuhp/1
idle_inject/1
migration/1
ksoftirqd/1
kdevtmpfs
inet_frag_wq
kauditd
khungtaskd
```

Tic Tac Toe

```
Tic Tac Toe

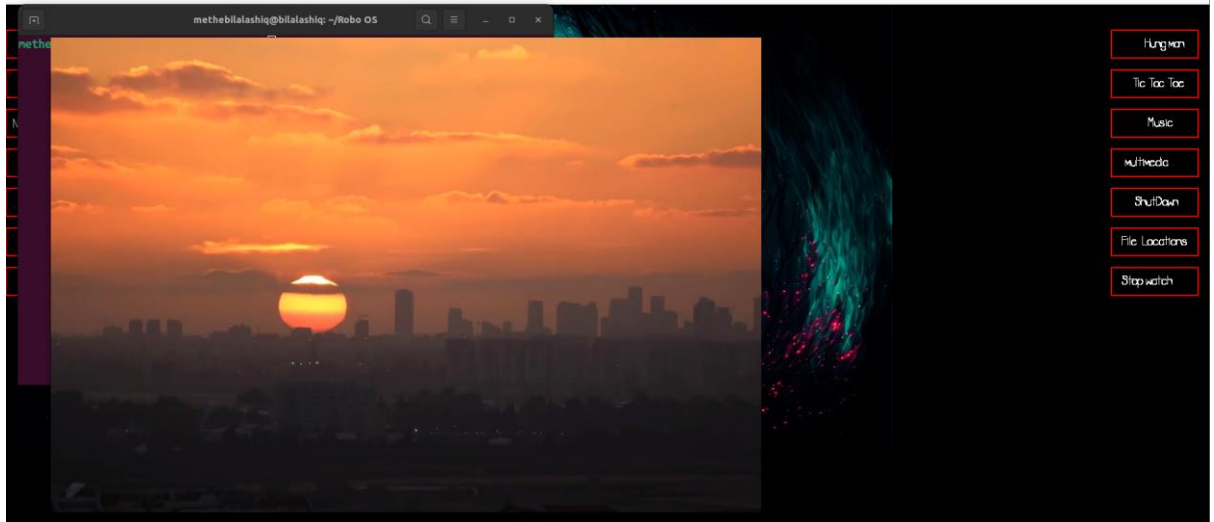
sh: 1: cls: not found
=====
                    TIC TAC TOE
=====
Player 1 (X) - Player 2 (O)

  1 | 2 | 3
  --|---|
  4 | 5 | 6
  --|---|
  7 | 8 | 9

Player 1, enter a number: █
```




Multi Media



Stop watch





Shut Down

```
methebilalashiq@bilalashiq: ~/Robo OS
INITIATING OPERATING SYSTEM SHUTDOWN

----- >>>> Terminating All the Running Processes <<<< -----
----- >>>> Releasing System Resources <<<< -----
----- >>>> Saving USER DATA <<<< -----
----- >>>> Stopping system services...
----- >>>> Flushing disk buffers... <<<< -----
----- >>>> Do not exit this window <<<< -----

S
    3
    2
    1
```